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INSTALLATION STANDARDS · TECHNICAL NOTE

Installed to standard, checked on site

Solar ESS technical note · October 2026 · IMC, EMT, conduit, enclosures, heat, terminals



NEW IN THAILAND

Two mandatory PV standards

TISI approved มอก. 62852 (PV DC connectors) and มอก. 61439 เล่ม 8 (PV DC combiner boxes) on 18 August 2026, expected in force from December 2026.

HOW WE WORK

EIT code + IEC 62548

Thai code rules follow the EIT wiring code and the EIT rooftop PV standard, which match the NEC on most points.

Numbers to remember

Conduit fill limit

1 cable	53 %
2 cables	31 %
3 or more cables	**40 %**

Max ambient for best life

Inverter (advised)	40 °C
Battery (best range top)	45 °C
Breaker calibration (IEC 60898)	30 °C

Enclosure IP rating

Outdoors (minimum)	IP65
Coastal / humid	**IP66**

Impact rating

IK08 (roof)	5 J
IK10 (public area)	**20 J**

Conduit

Type	Standard	Use it for	Check
IMC	มอก. 770, UL 1242	**Indoor, outdoor, wet, embedded**	Print, wall, support every 3 m
EMT	มอก. 770, UL 797	Indoor and dry only	Never exposed on roofs
RSC	มอก. 770, UL 6	**All locations, coastal, buried**	Hot-dip galvanising
Flexible metal	UL 1	Dry, short drops	Never the earth path

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Type	Standard	Use it for	Check
Liquid-tight flex	UL 360	Wet drops to inverter/battery	UV-rated glands
PVC / HDPE	Maker / TIS	UV-rated uPVC outdoors; HDPE buried	Expansion fittings on hot roofs

Enclosures, heat and terminals

Item	Requirement	Check
Combiner box	IEC 61439-8, IP65+, IK08+, DC rating ≥ cold Voc	Bottom entry, glands tight, blanks fitted
DC equipment	IEC 62548-1: DC isolator, fuses 1.5–2.4 × Isc, SPD > Voc	Load-break test, SPD window, labels
Inverter heat	250–300 mm clearance, shade, no attics	Watch for temperature derating
Battery heat	Best 10–45 °C; Solis IntelliHome not outdoors within 500 m of the sea	Check April temperatures
PV connectors	Same type and maker on both halves (มอก. 62852)	No mixed brands; maker's crimp tool
Lugs, ferrules, torque	IEC 61238-1-1; maker's torque	Die number, torque mark on every terminal

The 6 mistakes we see most

CONDUIT
EMT on the roof
Use IMC outdoors

UV
Yellow PVC in the sun
Cracks in 2–3 years

CONNECTORS
Mixed MC4 brands
Main cause of DC arcing

BOXES
Open or top entries
Water gets in

PROTECTION
AC breakers on DC
Cannot break a DC arc

HEAT
Inverter in sun or attic
Permanent derating

Specifications are from the manufacturers' published datasheets; star ratings are Solar ESS's own assessment. Check the current datasheet before ordering. For prices and a design for your site, book a consultation at www.solaress.org or LINE: solaress1.

Sources: IEC 62548-1:2023 · TISI PV standards · EIT rooftop PV standard summary